Reversed achievement score in high school for the youngest child

Current 2022/2023 enrollment defines the high school sample. One youngest child within this level is selected per household before subject-specific outcome exclusions. 6 minus the recorded response for responses 1 through 5. Higher scores indicate better achievement. Codes 6 and 99 and blanks are excluded separately for each subject. Father estimates are primary and mother estimates are secondary.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.619	0.043 [0.314] N=774	0.104 [0.311] N=615	0.095 [0.315] N=466	0.092 [0.338] N=324	0.203 [0.400] N=195
Social studies and life sciences	3.635	-0.038 [0.208] N=736	-0.003 [0.222] N=583	-0.016 [0.247] N=443	-0.053 [0.252] N=309	0.223 [0.266] N=189
Mathematics	3.365	-0.212 [0.353] N=762	-0.092 [0.364] N=604	-0.075 [0.381] N=457	-0.198 [0.425] N=317	-0.048 [0.538] N=191
Science	3.545	0.010 [0.339] N=737	-0.002 [0.337] N=585	-0.018 [0.356] N=450	-0.031 [0.394] N=314	0.077 [0.522] N=192
Foreign language	3.424	**-0.535* [0.312] N=752**	**-0.602** [0.293] N=598**	**-0.712** [0.309] N=453**	**-0.626* [0.360] N=315**	-0.257 [0.425] N=190
Culture arts and sports	3.818	-0.174 [0.495] N=697	-0.140 [0.474] N=547	-0.149 [0.452] N=417	0.018 [0.542] N=286	-0.048 [0.550] N=173
Vocational and technological	3.728	**-0.616** [0.269] N=548**	**-0.556* [0.291] N=434**	**-0.609** [0.295] N=324**	-0.509 [0.334] N=224	-0.213 [0.391] N=138

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.144 [0.339] N=758	0.220 [0.343] N=599	0.220 [0.357] N=450	0.206 [0.389] N=308	0.399 [0.498] N=179
Social studies and life sciences	-0.130 [0.261] N=720	-0.095 [0.279] N=567	-0.116 [0.313] N=427	-0.174 [0.317] N=293	0.220 [0.344] N=173
Mathematics	-0.145 [0.454] N=747	0.019 [0.464] N=589	0.075 [0.492] N=442	-0.063 [0.564] N=302	0.337 [0.760] N=176
Science	-0.186 [0.377] N=722	-0.168 [0.388] N=570	-0.187 [0.406] N=435	-0.182 [0.467] N=299	-0.007 [0.657] N=177
Foreign language	-0.565 [0.365] N=738	**-0.621* [0.340] N=584**	**-0.727** [0.361] N=439**	-0.599 [0.425] N=301	-0.006 [0.493] N=176
Culture arts and sports	-0.067 [0.635] N=683	0.049 [0.551] N=533	0.032 [0.530] N=403	0.308 [0.633] N=272	0.283 [0.657] N=159
Vocational and technological	**-0.925*** [0.290] N=537**	**-0.807*** [0.273] N=423**	**-0.911*** [0.291] N=313**	**-0.727* [0.376] N=213**	-0.297 [0.433] N=127

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.619	0.085 [0.283] N=774	0.165 [0.279] N=615	0.167 [0.281] N=466	0.119 [0.316] N=324	0.194 [0.391] N=195
Social studies and life sciences	3.635	-0.002 [0.189] N=736	0.039 [0.207] N=583	0.041 [0.230] N=443	-0.026 [0.244] N=309	0.219 [0.268] N=189
Mathematics	3.365	-0.191 [0.344] N=762	-0.070 [0.354] N=604	-0.040 [0.369] N=457	-0.191 [0.422] N=317	-0.051 [0.544] N=191
Science	3.545	0.050 [0.341] N=737	0.044 [0.339] N=585	0.030 [0.360] N=450	-0.015 [0.402] N=314	0.071 [0.544] N=192
Foreign language	3.424	**-0.505* [0.281] N=752**	**-0.554** [0.263] N=598**	**-0.659** [0.277] N=453**	**-0.627* [0.330] N=315**	-0.280 [0.412] N=190
Culture arts and sports	3.818	-0.164 [0.484] N=697	-0.134 [0.470] N=547	-0.147 [0.451] N=417	0.018 [0.535] N=286	-0.056 [0.529] N=173
Vocational and technological	3.728	**-0.613** [0.274] N=548**	**-0.555* [0.292] N=434**	**-0.606* [0.304] N=324**	-0.503 [0.349] N=224	-0.214 [0.395] N=138

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.152 [0.310] N=758	0.245 [0.316] N=599	0.256 [0.328] N=450	0.196 [0.379] N=308	0.366 [0.511] N=179
Social studies and life sciences	-0.124 [0.229] N=720	-0.078 [0.255] N=567	-0.089 [0.283] N=427	-0.175 [0.301] N=293	0.193 [0.353] N=173
Mathematics	-0.140 [0.445] N=747	0.029 [0.455] N=589	0.097 [0.481] N=442	-0.066 [0.565] N=302	0.315 [0.781] N=176
Science	-0.174 [0.358] N=722	-0.145 [0.374] N=570	-0.160 [0.394] N=435	-0.185 [0.466] N=299	-0.032 [0.674] N=177
Foreign language	**-0.559* [0.329] N=738**	**-0.596* [0.310] N=584**	**-0.691** [0.327] N=439**	-0.611 [0.397] N=301	-0.030 [0.486] N=176
Culture arts and sports	-0.072 [0.628] N=683	0.048 [0.549] N=533	0.031 [0.530] N=403	0.300 [0.631] N=272	0.252 [0.643] N=159
Vocational and technological	**-0.936*** [0.291] N=537**	**-0.812*** [0.276] N=423**	**-0.924*** [0.299] N=313**	**-0.752* [0.392] N=213**	-0.305 [0.445] N=127

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.619	0.130 [0.288] N=774	0.232 [0.273] N=615	0.276 [0.278] N=466	0.238 [0.290] N=324	0.414 [0.372] N=195
Social studies and life sciences	3.635	0.002 [0.199] N=736	0.053 [0.218] N=583	0.119 [0.249] N=443	0.079 [0.260] N=309	**0.463* [0.271] N=189**
Mathematics	3.365	-0.180 [0.352] N=762	-0.035 [0.364] N=604	0.000 [0.380] N=457	-0.096 [0.452] N=317	0.169 [0.605] N=191
Science	3.545	0.073 [0.329] N=737	0.094 [0.329] N=585	0.120 [0.345] N=450	0.108 [0.408] N=314	0.424 [0.541] N=192
Foreign language	3.424	**-0.549* [0.313] N=752**	**-0.586* [0.297] N=598**	**-0.603* [0.332] N=453**	-0.527 [0.329] N=315	-0.034 [0.425] N=190
Culture arts and sports	3.818	-0.224 [0.493] N=697	-0.223 [0.476] N=547	-0.236 [0.455] N=417	-0.032 [0.525] N=286	-0.228 [0.550] N=173
Vocational and technological	3.728	**-0.572* [0.315] N=548**	**-0.578* [0.341] N=434**	**-0.615* [0.359] N=324**	-0.416 [0.422] N=224	-0.183 [0.448] N=138

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.234 [0.308] N=758	0.352 [0.299] N=599	0.427 [0.318] N=450	0.377 [0.347] N=308	0.726 [0.481] N=179
Social studies and life sciences	-0.092 [0.250] N=720	-0.033 [0.279] N=567	0.043 [0.327] N=427	-0.021 [0.347] N=293	0.584 [0.391] N=173
Mathematics	-0.114 [0.463] N=747	0.082 [0.475] N=589	0.156 [0.507] N=442	0.050 [0.613] N=302	0.603 [0.867] N=176
Science	-0.136 [0.355] N=722	-0.080 [0.371] N=570	-0.057 [0.388] N=435	-0.042 [0.485] N=299	0.415 [0.722] N=177
Foreign language	-0.585 [0.365] N=738	**-0.613* [0.347] N=584**	-0.610 [0.384] N=439	-0.489 [0.377] N=301	0.188 [0.469] N=176
Culture arts and sports	-0.121 [0.640] N=683	-0.036 [0.561] N=533	-0.030 [0.529] N=403	0.304 [0.603] N=272	0.039 [0.686] N=159
Vocational and technological	**-0.821* [0.423] N=537**	**-0.787* [0.395] N=423**	**-0.870** [0.434] N=313**	-0.582 [0.558] N=213	-0.282 [0.564] N=127

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.619	0.093 [0.281] N=774	0.199 [0.273] N=615	0.220 [0.277] N=466	0.182 [0.307] N=324	0.362 [0.389] N=195
Social studies and life sciences	3.635	-0.039 [0.195] N=736	0.030 [0.212] N=583	0.074 [0.238] N=443	-0.006 [0.261] N=309	0.358 [0.289] N=189
Mathematics	3.365	-0.249 [0.347] N=762	-0.129 [0.353] N=604	-0.111 [0.366] N=457	-0.234 [0.453] N=317	0.038 [0.589] N=191
Science	3.545	0.018 [0.329] N=737	0.024 [0.324] N=585	0.032 [0.332] N=450	0.041 [0.403] N=314	0.274 [0.541] N=192
Foreign language	3.424	**-0.587** [0.291] N=752**	**-0.632** [0.276] N=598**	**-0.718** [0.296] N=453**	**-0.661** [0.307] N=315**	-0.198 [0.418] N=190
Culture arts and sports	3.818	-0.209 [0.472] N=697	-0.162 [0.458] N=547	-0.180 [0.448] N=417	-0.021 [0.532] N=286	-0.167 [0.567] N=173
Vocational and technological	3.728	-0.503 [0.319] N=548	-0.462 [0.328] N=434	-0.473 [0.376] N=324	-0.254 [0.435] N=224	-0.025 [0.490] N=138

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.153 [0.301] N=758	0.273 [0.297] N=599	0.308 [0.307] N=450	0.243 [0.355] N=308	0.475 [0.490] N=179
Social studies and life sciences	-0.167 [0.233] N=720	-0.098 [0.263] N=567	-0.058 [0.300] N=427	-0.168 [0.318] N=293	0.255 [0.383] N=173
Mathematics	-0.206 [0.453] N=747	-0.031 [0.458] N=589	0.028 [0.483] N=442	-0.081 [0.602] N=302	0.458 [0.834] N=176
Science	-0.198 [0.349] N=722	-0.162 [0.361] N=570	-0.155 [0.368] N=435	-0.134 [0.472] N=299	0.149 [0.699] N=177
Foreign language	**-0.629* [0.340] N=738**	**-0.668** [0.323] N=584**	**-0.723** [0.343] N=439**	**-0.650* [0.361] N=301**	0.007 [0.495] N=176
Culture arts and sports	-0.114 [0.607] N=683	0.020 [0.526] N=533	0.003 [0.518] N=403	0.250 [0.624] N=272	0.102 [0.722] N=159
Vocational and technological	**-0.807** [0.372] N=537**	**-0.715** [0.335] N=423**	**-0.763* [0.401] N=313**	-0.487 [0.506] N=213	-0.091 [0.605] N=127

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.617	0.096 [0.281] N=734	0.151 [0.274] N=584	0.143 [0.287] N=445	0.134 [0.319] N=312	0.205 [0.409] N=187
Social studies and life sciences	3.635	0.013 [0.186] N=696	0.049 [0.207] N=552	0.029 [0.235] N=422	-0.003 [0.240] N=297	0.234 [0.266] N=181
Mathematics	3.360	-0.184 [0.343] N=722	-0.075 [0.355] N=573	-0.028 [0.369] N=436	-0.164 [0.418] N=305	-0.103 [0.563] N=183
Science	3.540	0.037 [0.340] N=698	0.030 [0.340] N=555	0.027 [0.359] N=429	0.010 [0.398] N=302	0.042 [0.569] N=184
Foreign language	3.424	**-0.518* [0.289] N=713**	**-0.562** [0.268] N=567**	**-0.640** [0.290] N=432**	**-0.591* [0.315] N=303**	-0.192 [0.456] N=182
Culture arts and sports	3.815	-0.160 [0.479] N=662	-0.152 [0.469] N=520	-0.164 [0.452] N=398	0.004 [0.543] N=275	-0.092 [0.546] N=165
Vocational and technological	3.730	**-0.621** [0.272] N=518**	**-0.587* [0.295] N=410**	**-0.580* [0.305] N=309**	-0.548 [0.361] N=214	-0.284 [0.416] N=130

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.179 [0.304] N=718	0.244 [0.304] N=568	0.234 [0.331] N=429	0.227 [0.376] N=296	0.400 [0.544] N=171
Social studies and life sciences	-0.097 [0.226] N=680	-0.057 [0.256] N=536	-0.096 [0.298] N=406	-0.139 [0.303] N=281	0.217 [0.375] N=165
Mathematics	-0.125 [0.449] N=707	0.029 [0.460] N=558	0.115 [0.485] N=421	-0.027 [0.569] N=290	0.254 [0.832] N=168
Science	-0.182 [0.360] N=683	-0.156 [0.381] N=540	-0.163 [0.401] N=414	-0.152 [0.472] N=287	-0.080 [0.726] N=169
Foreign language	**-0.564* [0.338] N=699**	**-0.598* [0.317] N=553**	**-0.665* [0.340] N=418**	-0.575 [0.367] N=289	0.039 [0.525] N=168
Culture arts and sports	-0.055 [0.612] N=648	0.035 [0.538] N=506	0.022 [0.518] N=384	0.290 [0.632] N=261	0.211 [0.675] N=151
Vocational and technological	**-0.943*** [0.271] N=507**	**-0.849*** [0.264] N=399**	**-0.839** [0.319] N=298**	-0.741 [0.458] N=203	-0.313 [0.545] N=119

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.617	0.097 [0.281] N=734	0.158 [0.274] N=584	0.135 [0.284] N=445	0.135 [0.318] N=312	0.249 [0.407] N=187
Social studies and life sciences	3.635	0.005 [0.191] N=696	0.044 [0.209] N=552	0.018 [0.234] N=422	-0.013 [0.244] N=297	0.268 [0.275] N=181
Mathematics	3.360	-0.180 [0.347] N=722	-0.066 [0.358] N=573	-0.022 [0.372] N=436	-0.161 [0.427] N=305	-0.033 [0.553] N=183
Science	3.540	0.046 [0.337] N=698	0.041 [0.336] N=555	0.021 [0.356] N=429	0.005 [0.397] N=302	0.080 [0.556] N=184
Foreign language	3.424	**-0.491* [0.287] N=713**	**-0.529** [0.265] N=567**	**-0.623** [0.276] N=432**	**-0.573* [0.309] N=303**	-0.162 [0.434] N=182
Culture arts and sports	3.815	-0.152 [0.483] N=662	-0.140 [0.471] N=520	-0.157 [0.452] N=398	0.020 [0.542] N=275	-0.068 [0.542] N=165
Vocational and technological	3.730	**-0.596** [0.276] N=518**	**-0.574* [0.300] N=410**	**-0.589* [0.306] N=309**	-0.492 [0.358] N=214	-0.225 [0.417] N=130

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.182 [0.303] N=718	0.252 [0.305] N=568	0.228 [0.328] N=429	0.225 [0.373] N=296	0.444 [0.524] N=171
Social studies and life sciences	-0.105 [0.232] N=680	-0.063 [0.260] N=536	-0.106 [0.296] N=406	-0.154 [0.305] N=281	0.251 [0.369] N=165
Mathematics	-0.120 [0.452] N=707	0.039 [0.462] N=558	0.120 [0.487] N=421	-0.028 [0.574] N=290	0.339 [0.802] N=168
Science	-0.169 [0.358] N=683	-0.142 [0.377] N=540	-0.166 [0.398] N=414	-0.159 [0.469] N=287	-0.029 [0.697] N=169
Foreign language	-0.529 [0.334] N=699	**-0.555* [0.308] N=553**	**-0.638** [0.320] N=418**	-0.544 [0.354] N=289	0.092 [0.493] N=168
Culture arts and sports	-0.047 [0.622] N=648	0.053 [0.545] N=506	0.034 [0.522] N=384	0.319 [0.632] N=261	0.243 [0.671] N=151
Vocational and technological	**-0.920*** [0.283] N=507**	**-0.847*** [0.272] N=399**	**-0.872*** [0.311] N=298**	-0.700 [0.423] N=203	-0.282 [0.504] N=119

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.643	-0.018 [0.084] N=1,468	-0.037 [0.087] N=1,314	-0.017 [0.089] N=1,123	0.002 [0.097] N=935	0.055 [0.106] N=749
Social studies and life sciences	3.658	-0.096 [0.095] N=1,397	-0.100 [0.097] N=1,252	-0.084 [0.098] N=1,069	-0.054 [0.105] N=891	-0.013 [0.110] N=713
Mathematics	3.369	0.029 [0.097] N=1,456	0.002 [0.096] N=1,302	0.022 [0.098] N=1,112	0.061 [0.101] N=926	0.082 [0.104] N=739
Science	3.559	0.078 [0.088] N=1,397	0.087 [0.090] N=1,249	0.080 [0.094] N=1,069	0.131 [0.103] N=889	**0.190* [0.108] N=713**
Foreign language	3.447	0.094 [0.106] N=1,433	0.073 [0.111] N=1,282	0.102 [0.117] N=1,093	0.107 [0.129] N=913	0.131 [0.138] N=731
Culture arts and sports	3.865	-0.001 [0.083] N=1,332	-0.002 [0.087] N=1,189	0.036 [0.090] N=1,014	0.075 [0.100] N=844	0.039 [0.106] N=679
Vocational and technological	3.749	0.062 [0.079] N=989	0.045 [0.079] N=882	0.046 [0.082] N=758	-0.016 [0.090] N=642	0.063 [0.103] N=522

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.103 [0.132] N=1,261	-0.133 [0.136] N=1,107	-0.089 [0.142] N=916	-0.035 [0.171] N=728	0.157 [0.212] N=542
Social studies and life sciences	-0.110 [0.140] N=1,200	-0.109 [0.142] N=1,055	-0.080 [0.146] N=872	-0.007 [0.170] N=694	0.171 [0.203] N=516
Mathematics	0.139 [0.161] N=1,253	0.098 [0.165] N=1,099	0.158 [0.174] N=909	0.257 [0.195] N=723	**0.544** [0.223] N=536**
Science	0.068 [0.143] N=1,201	0.093 [0.147] N=1,053	0.098 [0.152] N=873	0.225 [0.176] N=693	**0.551*** [0.204] N=517**
Foreign language	0.043 [0.174] N=1,233	0.032 [0.178] N=1,082	0.099 [0.189] N=893	0.111 [0.238] N=713	0.224 [0.257] N=531
Culture arts and sports	0.102 [0.128] N=1,147	0.105 [0.129] N=1,004	0.170 [0.135] N=829	0.253 [0.167] N=659	0.137 [0.190] N=494
Vocational and technological	0.237 [0.149] N=841	0.231 [0.149] N=734	0.247 [0.155] N=610	0.131 [0.192] N=494	0.434 [0.276] N=374

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.643	-0.001 [0.086] N=1,468	-0.012 [0.087] N=1,314	0.005 [0.089] N=1,123	0.017 [0.098] N=935	0.069 [0.105] N=749
Social studies and life sciences	3.658	-0.082 [0.096] N=1,397	-0.079 [0.098] N=1,252	-0.068 [0.099] N=1,069	-0.040 [0.105] N=891	-0.003 [0.111] N=713
Mathematics	3.369	0.036 [0.098] N=1,456	0.012 [0.097] N=1,302	0.031 [0.099] N=1,112	0.064 [0.101] N=926	0.087 [0.105] N=739
Science	3.559	0.083 [0.088] N=1,397	0.099 [0.090] N=1,249	0.087 [0.093] N=1,069	0.135 [0.102] N=889	**0.192* [0.107] N=713**
Foreign language	3.447	0.108 [0.108] N=1,433	0.094 [0.113] N=1,282	0.121 [0.119] N=1,093	0.118 [0.130] N=913	0.140 [0.139] N=731
Culture arts and sports	3.865	0.005 [0.082] N=1,332	0.008 [0.086] N=1,189	0.047 [0.088] N=1,014	0.082 [0.098] N=844	0.046 [0.102] N=679
Vocational and technological	3.749	0.062 [0.078] N=989	0.048 [0.079] N=882	0.051 [0.082] N=758	-0.017 [0.088] N=642	0.067 [0.101] N=522

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.076 [0.130] N=1,261	-0.096 [0.134] N=1,107	-0.060 [0.140] N=916	-0.024 [0.173] N=728	0.159 [0.204] N=542
Social studies and life sciences	-0.090 [0.136] N=1,200	-0.077 [0.139] N=1,055	-0.059 [0.142] N=872	0.006 [0.169] N=694	0.171 [0.194] N=516
Mathematics	0.149 [0.162] N=1,253	0.112 [0.166] N=1,099	0.167 [0.175] N=909	0.256 [0.195] N=723	**0.543** [0.223] N=536**
Science	0.081 [0.143] N=1,201	0.116 [0.146] N=1,053	0.113 [0.153] N=873	0.230 [0.178] N=693	**0.550*** [0.203] N=517**
Foreign language	0.067 [0.173] N=1,233	0.068 [0.178] N=1,082	0.127 [0.189] N=893	0.117 [0.240] N=713	0.220 [0.243] N=531
Culture arts and sports	0.110 [0.128] N=1,147	0.116 [0.129] N=1,004	0.179 [0.134] N=829	0.256 [0.167] N=659	0.127 [0.187] N=494
Vocational and technological	0.240 [0.149] N=841	0.238 [0.149] N=734	0.255 [0.155] N=610	0.129 [0.192] N=494	0.437 [0.275] N=374

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.643	-0.006 [0.084] N=1,468	-0.021 [0.086] N=1,314	0.000 [0.088] N=1,123	0.011 [0.095] N=935	0.050 [0.106] N=749
Social studies and life sciences	3.658	-0.085 [0.096] N=1,397	-0.084 [0.097] N=1,252	-0.071 [0.099] N=1,069	-0.046 [0.104] N=891	-0.020 [0.112] N=713
Mathematics	3.369	0.029 [0.098] N=1,456	0.005 [0.097] N=1,302	0.021 [0.098] N=1,112	0.056 [0.101] N=926	0.072 [0.106] N=739
Science	3.559	0.080 [0.087] N=1,397	0.096 [0.089] N=1,249	0.084 [0.092] N=1,069	0.131 [0.101] N=889	**0.179* [0.106] N=713**
Foreign language	3.447	0.104 [0.104] N=1,433	0.083 [0.109] N=1,282	0.112 [0.113] N=1,093	0.120 [0.124] N=913	0.125 [0.136] N=731
Culture arts and sports	3.865	0.010 [0.080] N=1,332	0.012 [0.084] N=1,189	0.051 [0.086] N=1,014	0.088 [0.096] N=844	0.048 [0.101] N=679
Vocational and technological	3.749	0.061 [0.078] N=989	0.039 [0.078] N=882	0.044 [0.082] N=758	-0.018 [0.090] N=642	0.059 [0.104] N=522

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.075 [0.123] N=1,261	-0.100 [0.128] N=1,107	-0.057 [0.131] N=916	-0.000 [0.161] N=728	0.157 [0.204] N=542
Social studies and life sciences	-0.084 [0.131] N=1,200	-0.075 [0.133] N=1,055	-0.049 [0.135] N=872	0.031 [0.158] N=694	0.169 [0.191] N=516
Mathematics	0.139 [0.164] N=1,253	0.098 [0.167] N=1,099	0.152 [0.172] N=909	0.250 [0.192] N=723	**0.520** [0.226] N=536**
Science	0.083 [0.139] N=1,201	0.114 [0.143] N=1,053	0.119 [0.147] N=873	0.245 [0.171] N=693	**0.546*** [0.204] N=517**
Foreign language	0.082 [0.162] N=1,233	0.072 [0.167] N=1,082	0.135 [0.175] N=893	0.153 [0.224] N=713	0.226 [0.241] N=531
Culture arts and sports	0.114 [0.127] N=1,147	0.117 [0.129] N=1,004	0.183 [0.136] N=829	**0.279* [0.165] N=659**	0.154 [0.191] N=494
Vocational and technological	0.222 [0.148] N=841	0.212 [0.148] N=734	0.229 [0.152] N=610	0.134 [0.190] N=494	0.401 [0.279] N=374

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.643	-0.009 [0.086] N=1,468	-0.019 [0.087] N=1,314	0.000 [0.090] N=1,123	0.015 [0.096] N=935	0.059 [0.107] N=749
Social studies and life sciences	3.658	-0.089 [0.096] N=1,397	-0.084 [0.097] N=1,252	-0.072 [0.098] N=1,069	-0.040 [0.104] N=891	-0.010 [0.113] N=713
Mathematics	3.369	0.035 [0.098] N=1,456	0.012 [0.098] N=1,302	0.029 [0.099] N=1,112	0.063 [0.101] N=926	0.079 [0.107] N=739
Science	3.559	0.080 [0.089] N=1,397	0.098 [0.091] N=1,249	0.086 [0.094] N=1,069	0.139 [0.102] N=889	**0.185* [0.110] N=713**
Foreign language	3.447	0.113 [0.104] N=1,433	0.098 [0.110] N=1,282	0.128 [0.114] N=1,093	0.138 [0.125] N=913	0.145 [0.137] N=731
Culture arts and sports	3.865	-0.003 [0.079] N=1,332	0.002 [0.083] N=1,189	0.043 [0.084] N=1,014	0.086 [0.093] N=844	0.038 [0.096] N=679
Vocational and technological	3.749	0.048 [0.075] N=989	0.037 [0.075] N=882	0.039 [0.078] N=758	-0.013 [0.084] N=642	0.053 [0.095] N=522

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.060 [0.123] N=1,261	-0.074 [0.128] N=1,107	-0.037 [0.134] N=916	0.015 [0.163] N=728	0.197 [0.203] N=542
Social studies and life sciences	-0.077 [0.127] N=1,200	-0.058 [0.129] N=1,055	-0.042 [0.132] N=872	0.032 [0.156] N=694	0.205 [0.180] N=516
Mathematics	0.161 [0.160] N=1,253	0.125 [0.163] N=1,099	0.181 [0.171] N=909	0.270 [0.189] N=723	**0.553** [0.221] N=536**
Science	0.091 [0.142] N=1,201	0.130 [0.144] N=1,053	0.126 [0.150] N=873	0.255 [0.173] N=693	**0.578*** [0.207] N=517**
Foreign language	0.105 [0.162] N=1,233	0.110 [0.166] N=1,082	0.173 [0.173] N=893	0.190 [0.222] N=713	0.263 [0.228] N=531
Culture arts and sports	0.124 [0.126] N=1,147	0.142 [0.127] N=1,004	0.210 [0.135] N=829	**0.321* [0.163] N=659**	0.195 [0.189] N=494
Vocational and technological	**0.237* [0.136] N=841**	**0.248* [0.136] N=734**	**0.265* [0.142] N=610**	0.194 [0.181] N=494	**0.436* [0.258] N=374**

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.638	0.026 [0.082] N=1,391	0.015 [0.083] N=1,247	0.032 [0.086] N=1,067	0.054 [0.094] N=890	0.059 [0.108] N=712
Social studies and life sciences	3.654	-0.036 [0.099] N=1,321	-0.031 [0.101] N=1,185	-0.021 [0.102] N=1,013	0.015 [0.107] N=846	0.008 [0.118] N=676
Mathematics	3.360	0.068 [0.097] N=1,379	0.050 [0.097] N=1,235	0.065 [0.098] N=1,056	0.106 [0.100] N=881	0.087 [0.107] N=702
Science	3.549	0.133 [0.087] N=1,323	0.147 [0.089] N=1,183	0.130 [0.092] N=1,013	**0.187* [0.100] N=844**	**0.199* [0.111] N=676**
Foreign language	3.441	0.143 [0.112] N=1,357	0.129 [0.118] N=1,216	0.168 [0.122] N=1,038	0.157 [0.133] N=868	0.128 [0.143] N=694
Culture arts and sports	3.861	-0.000 [0.081] N=1,270	0.004 [0.085] N=1,135	0.048 [0.088] N=969	0.083 [0.098] N=808	0.054 [0.103] N=649
Vocational and technological	3.750	0.055 [0.083] N=944	0.036 [0.084] N=840	0.045 [0.088] N=724	-0.024 [0.095] N=614	0.051 [0.105] N=500

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.002 [0.123] N=1,193	-0.024 [0.126] N=1,049	0.017 [0.129] N=869	0.077 [0.150] N=692	0.103 [0.197] N=514
Social studies and life sciences	-0.017 [0.132] N=1,133	-0.001 [0.135] N=997	0.029 [0.137] N=825	0.121 [0.154] N=658	0.132 [0.187] N=488
Mathematics	0.206 [0.165] N=1,185	0.179 [0.170] N=1,041	0.234 [0.174] N=862	**0.352* [0.187] N=687**	**0.506** [0.234] N=508**
Science	0.156 [0.139] N=1,136	0.193 [0.142] N=996	0.187 [0.144] N=826	**0.334** [0.157] N=657**	**0.522** [0.204] N=489**
Foreign language	0.136 [0.163] N=1,166	0.133 [0.169] N=1,025	0.219 [0.177] N=847	0.207 [0.223] N=677	0.165 [0.237] N=503
Culture arts and sports	0.123 [0.136] N=1,093	0.139 [0.137] N=958	0.215 [0.144] N=792	0.285 [0.177] N=631	0.151 [0.196] N=472
Vocational and technological	0.256 [0.159] N=801	0.251 [0.162] N=697	**0.290* [0.170] N=581**	0.145 [0.207] N=471	0.404 [0.293] N=357

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.638	0.026 [0.083] N=1,391	0.014 [0.085] N=1,247	0.033 [0.087] N=1,067	0.055 [0.094] N=890	0.062 [0.107] N=712
Social studies and life sciences	3.654	-0.042 [0.098] N=1,321	-0.038 [0.100] N=1,185	-0.028 [0.101] N=1,013	0.007 [0.105] N=846	0.004 [0.114] N=676
Mathematics	3.360	0.060 [0.097] N=1,379	0.041 [0.097] N=1,235	0.060 [0.097] N=1,056	0.101 [0.099] N=881	0.087 [0.106] N=702
Science	3.549	0.125 [0.088] N=1,323	0.139 [0.090] N=1,183	0.124 [0.092] N=1,013	**0.179* [0.100] N=844**	**0.194* [0.110] N=676**
Foreign language	3.441	0.143 [0.116] N=1,357	0.131 [0.121] N=1,216	0.172 [0.125] N=1,038	0.165 [0.137] N=868	0.140 [0.146] N=694
Culture arts and sports	3.861	0.009 [0.084] N=1,270	0.011 [0.088] N=1,135	0.057 [0.090] N=969	0.088 [0.101] N=808	0.059 [0.104] N=649
Vocational and technological	3.750	0.056 [0.083] N=944	0.036 [0.084] N=840	0.046 [0.088] N=724	-0.023 [0.095] N=614	0.054 [0.105] N=500

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.007 [0.123] N=1,193	-0.029 [0.126] N=1,049	0.015 [0.129] N=869	0.080 [0.151] N=692	0.125 [0.198] N=514
Social studies and life sciences	-0.016 [0.131] N=1,133	-0.003 [0.134] N=997	0.027 [0.135] N=825	0.111 [0.152] N=658	0.158 [0.187] N=488
Mathematics	0.194 [0.161] N=1,185	0.168 [0.166] N=1,041	0.235 [0.171] N=862	**0.356* [0.180] N=687**	**0.516** [0.227] N=508**
Science	0.149 [0.138] N=1,136	0.186 [0.141] N=996	0.183 [0.143] N=826	**0.333** [0.154] N=657**	**0.528*** [0.198] N=489**
Foreign language	0.113 [0.165] N=1,166	0.113 [0.170] N=1,025	0.202 [0.177] N=847	0.189 [0.223] N=677	0.161 [0.237] N=503
Culture arts and sports	0.132 [0.132] N=1,093	0.144 [0.134] N=958	0.222 [0.139] N=792	0.285 [0.175] N=631	0.159 [0.200] N=472
Vocational and technological	0.256 [0.156] N=801	0.252 [0.158] N=697	**0.292* [0.167] N=581**	0.150 [0.207] N=471	0.436 [0.290] N=357

Notes: Cells report the reform eligibility coefficient, parental birth-month-clustered standard error in brackets, and N. All models use exact child-month fixed effects calculated on 16 December 2022, split linear trends at January 1987, and FAKTOR_FERT child survey weights. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add categorical aid count and any aid. Aid specifications require valid yes/no answers to all six household aid questions. Income and aid controls are separate alternatives. Columns use 8- to 4-year bandwidths. Donuts omit running months -12 through +11. No child-age interval restriction is imposed; complete birth dates are needed for sibling selection and monthly controls. P is the unweighted positive-case count in the no-donut 8-year sample. NA denotes a failed or zero-standard-error fit. These are caregiver assessments. School-level and subject-response selection can affect interpretation. *** p<0.01, ** p<0.05, * p<0.10.